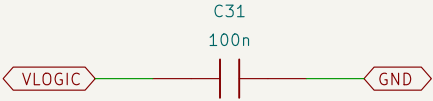
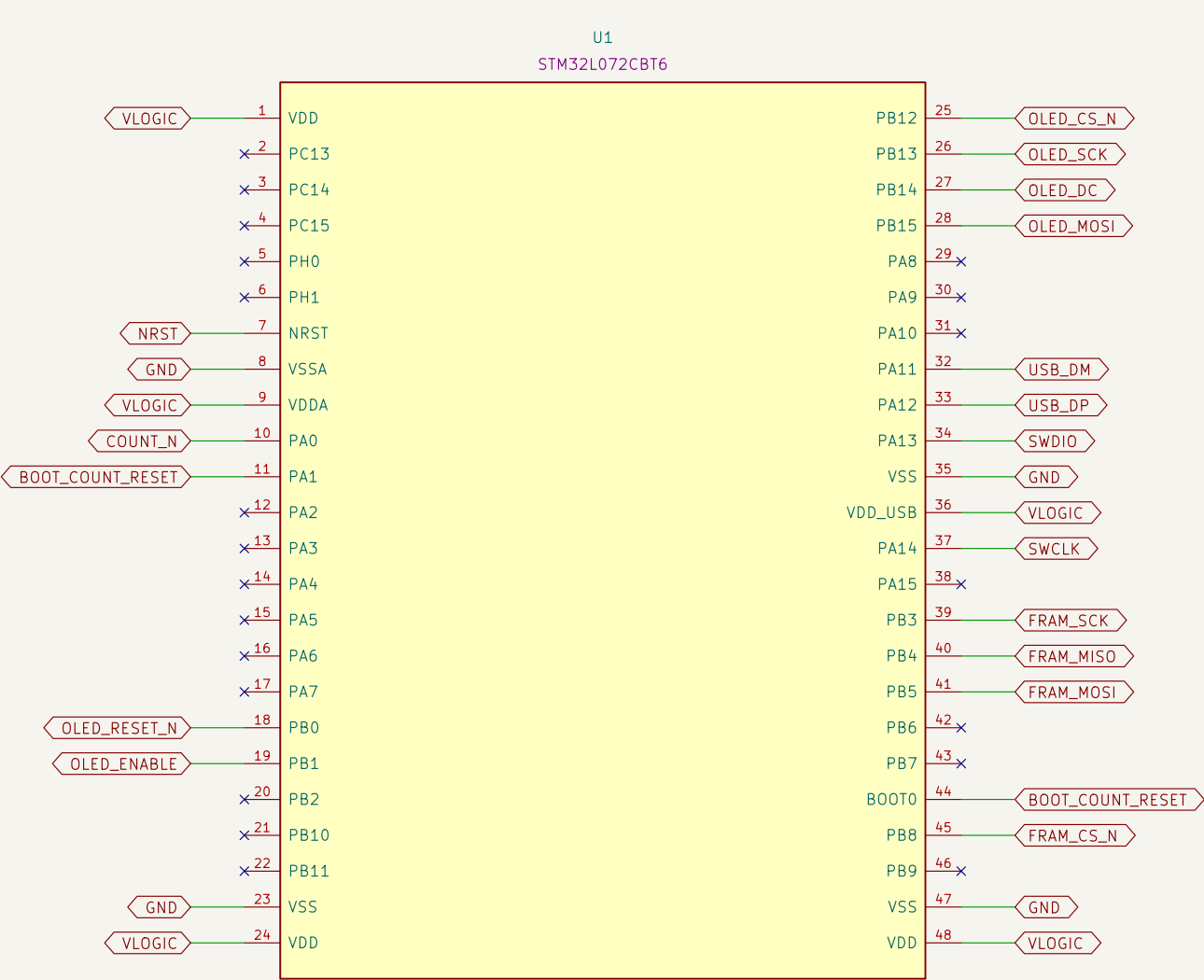
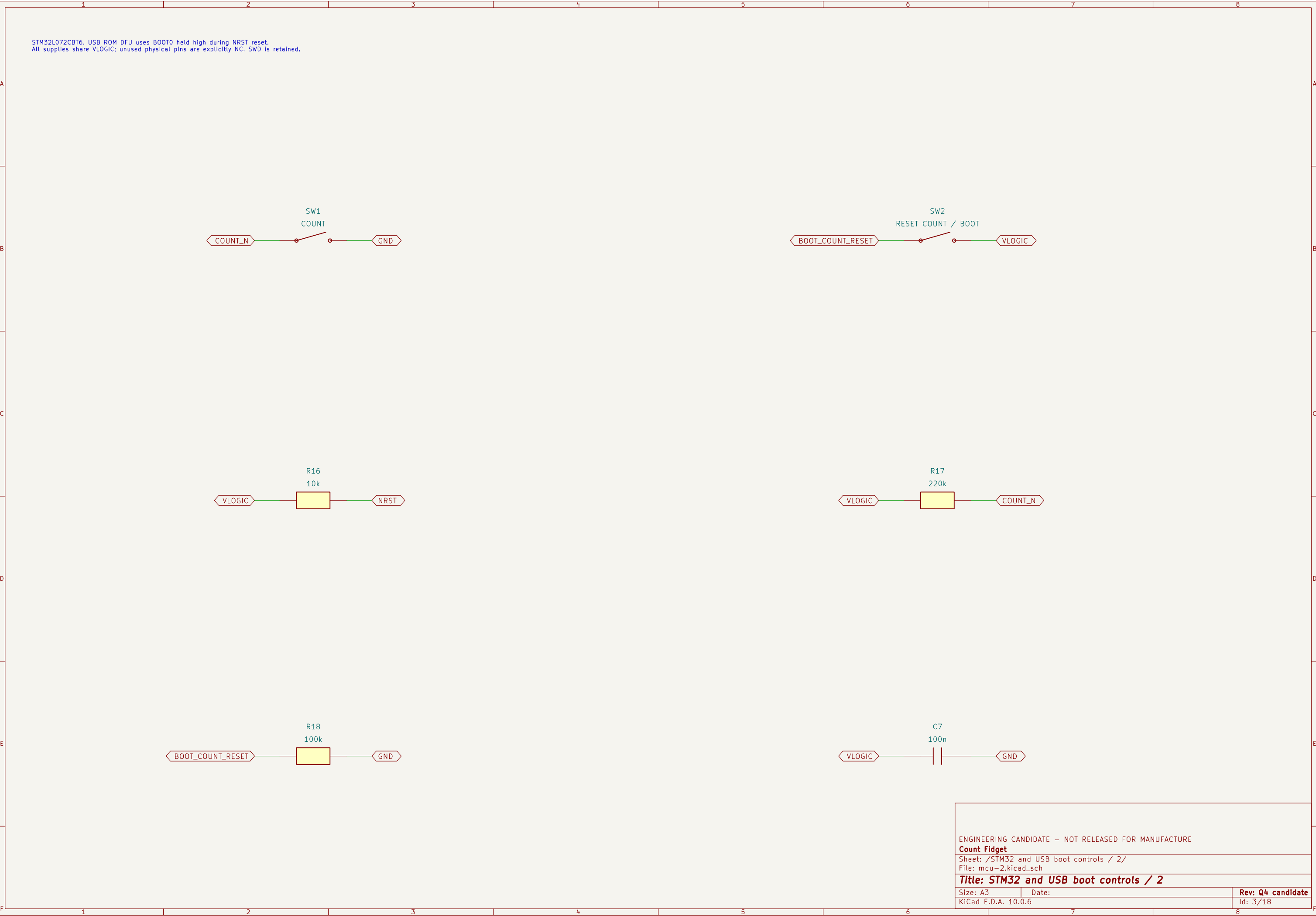


	1	2	3	4	5	6	7	8	
A	Q4 schematic index – 70 fitted PCB parts plus non-fitted test lands and mounts. Global net labels connect the functional sheets. Crosses explicitly mark unused physical pins. Q1 submitted files remain frozen. This candidate still requires ERC, layout, first-article and safety qualification.								A
	STM32 and USB boot controls		STM32 and USB boot controls / 2			STM32 and USB boot controls / 3			
	U1 / C31 / C32 / C33 / C34		SW1 / SW2 / R16 / R17 / R18 / C7			C6 / C10 / C11 / C12 / TP1 / TP2			
B	File: mcu-1.kicad_sch STM32 and USB boot controls / 4		File: mcu-2.kicad_sch STM32 and USB boot controls / 5			File: mcu-3.kicad_sch STM32 and USB boot controls / 6			B
	TP3 / TP4 / TP5 / TP6 / TP7 / TP8		TP9 / TP10 / H1 / H2 / SW3 / TP11			TP12 / C35 / H3 / H4			
	File: mcu-4.kicad_sch SPI OLED module and power control / 1		File: mcu-5.kicad_sch SPI OLED module and power control / 2			File: mcu-6.kicad_sch Independent SPI retained-count memory / 1			
C	DS1 / U6 / R24 / R25 / C20 / R29		C28			U8 / R32 / R33 / C30 / C36			C
	File: display-1.kicad_sch USB, charging and common supply / 1		File: display-2.kicad_sch USB, charging and common supply / 2			File: memory-1.kicad_sch USB, charging and common supply / 3			
	J1 / U2 / U3 / D1 / R1 / R2		R3 / R4 / R5 / R6 / R19 / C1			C2 / C3 / C5 / C13 / U7 / R30			
D	File: usb-power-1.kicad_sch USB, charging and common supply / 4		File: usb-power-2.kicad_sch Battery connection and protection / 1			File: usb-power-3.kicad_sch Battery connection and protection / 2			D
	R31 / C29 / C38 / C39 / D2		J2 / U4 / Q1 / Q2 / R7 / R8			R9 / C4			
	File: usb-power-4.kicad_sch Independent charge-temperature window / 1		File: battery-1.kicad_sch Independent charge-temperature window / 2			File: battery-2.kicad_sch			
E	U5 / Q3 / R10 / R11 / R12 / R13		R14 / R15 / C9 / Q4						E
	File: thermal-1.kicad_sch		File: thermal-2.kicad_sch						
F	1	2	3	4	5	6	7	8	F

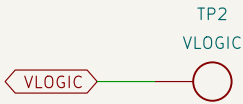
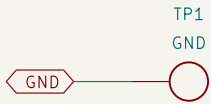
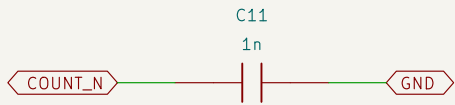
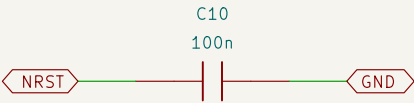
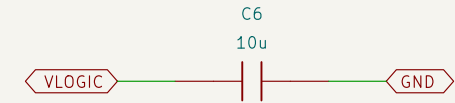
ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE		
Count Fidget		
Sheet: /		
File: click-counter-Q4.kicad_sch		
Title: Count Fidget – Q4 engineering candidate		
Size: A3	Date:	Rev: Q4 candidate
KiCad E.D.A. 10.0.6		Id: 1/18

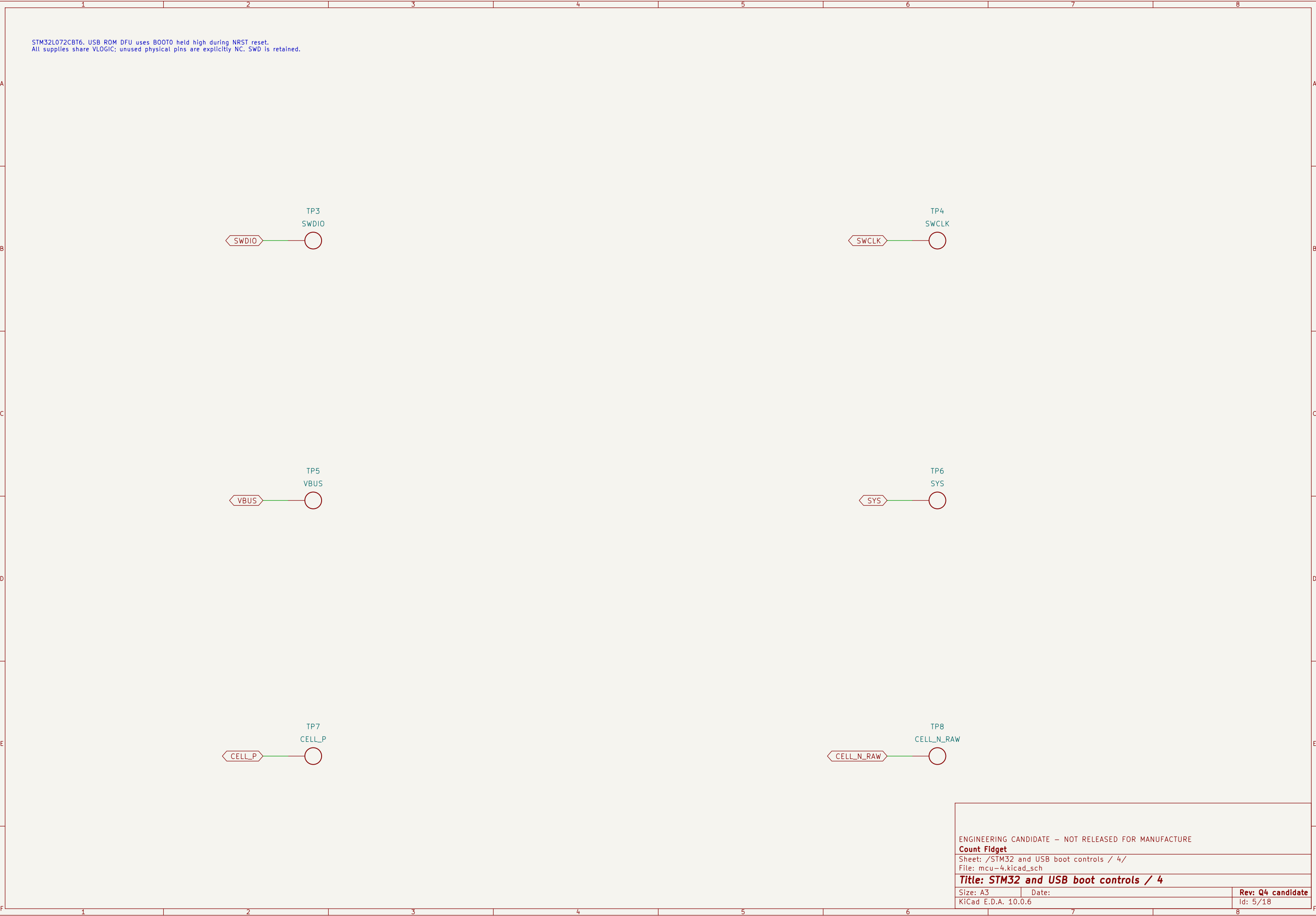
STM32L072CBT6. USB ROM DFU uses BOOT0 held high during NRST reset.
All supplies share VLOGIC; unused physical pins are explicitly NC. SWD is retained.

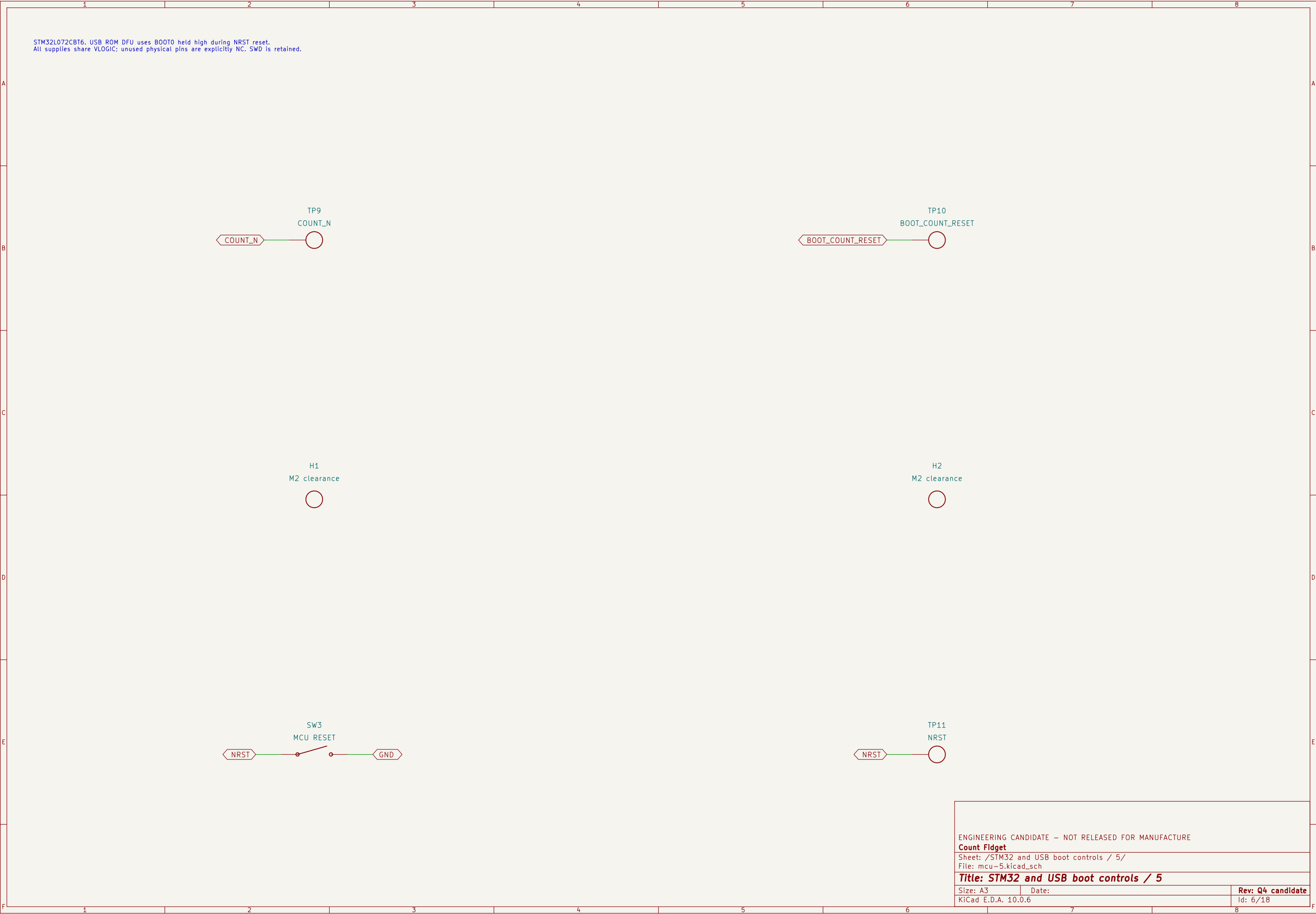




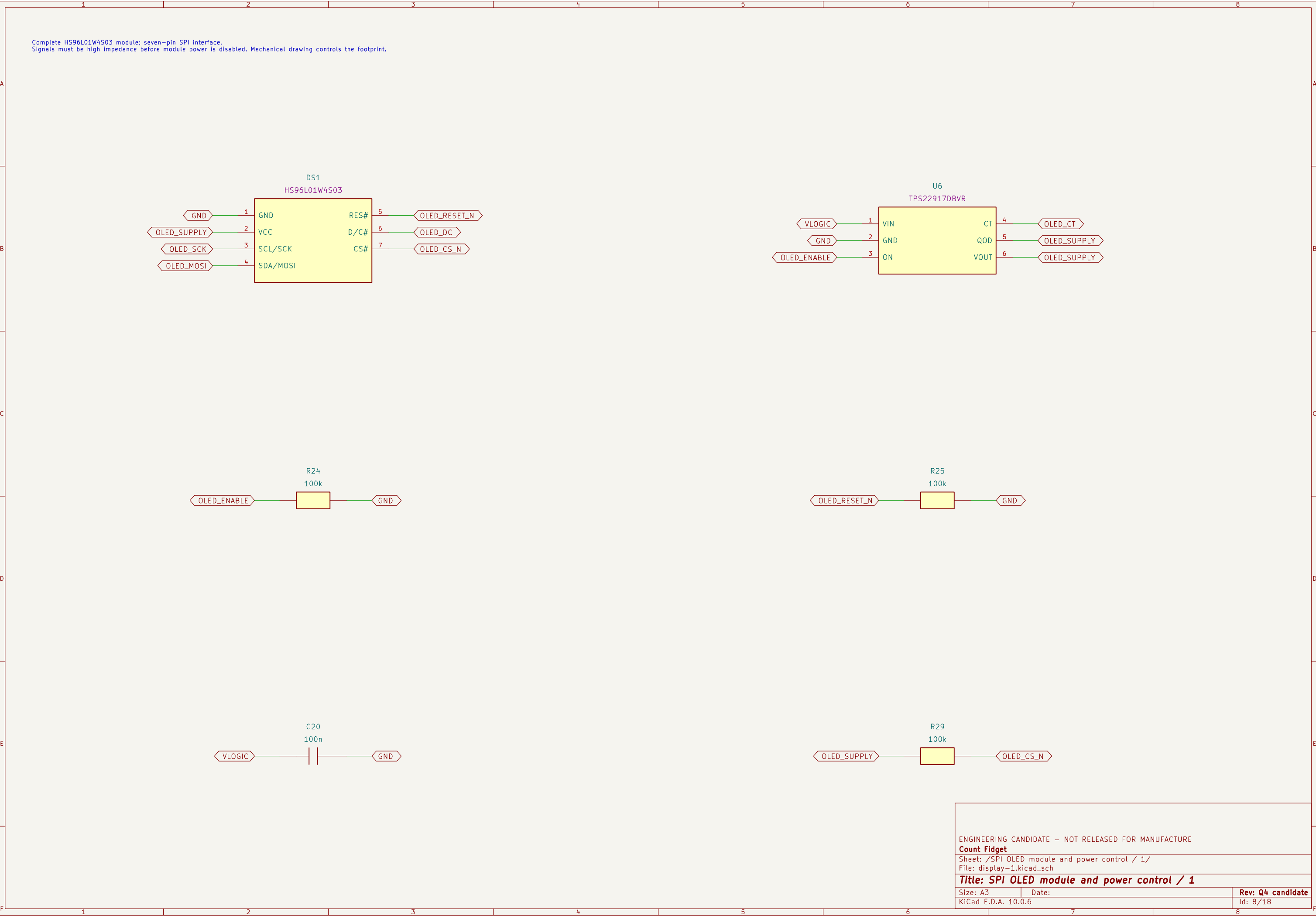
STM32L072CBT6. USB ROM DFU uses BOOT0 held high during NRST reset.
All supplies share VLOGIC; unused physical pins are explicitly NC. SWD is retained.



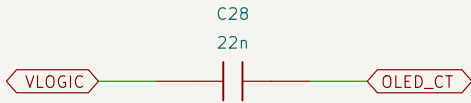


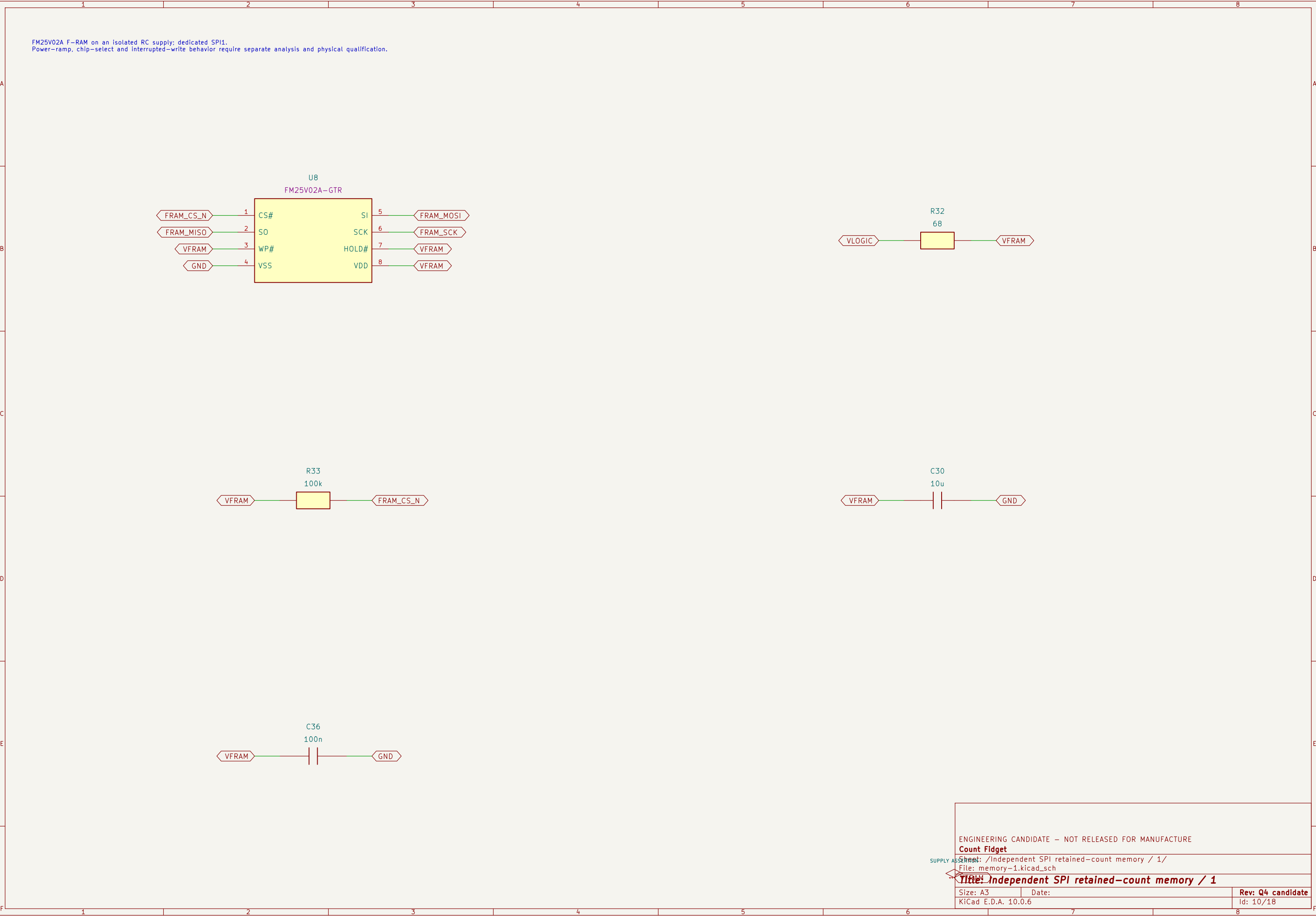


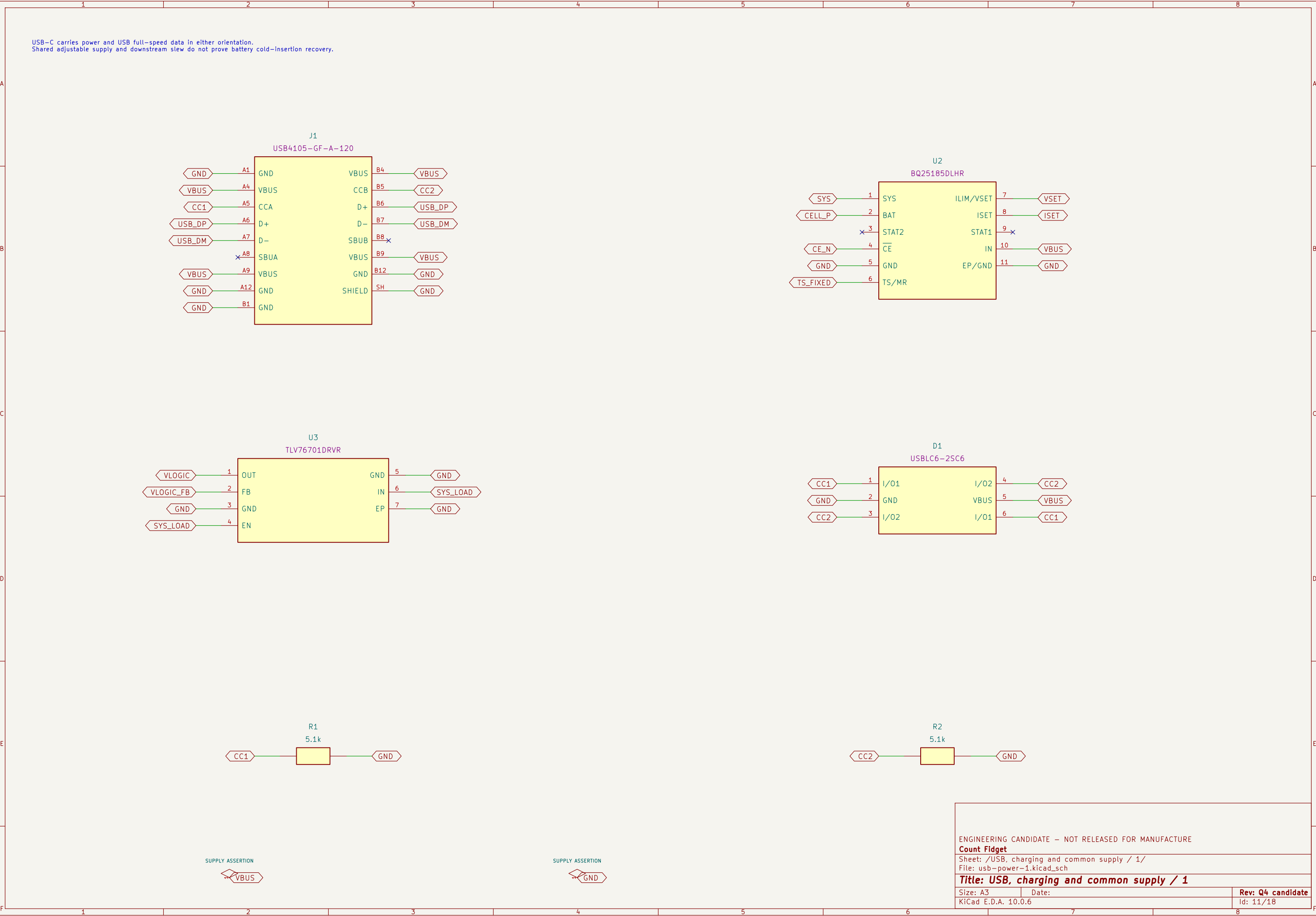


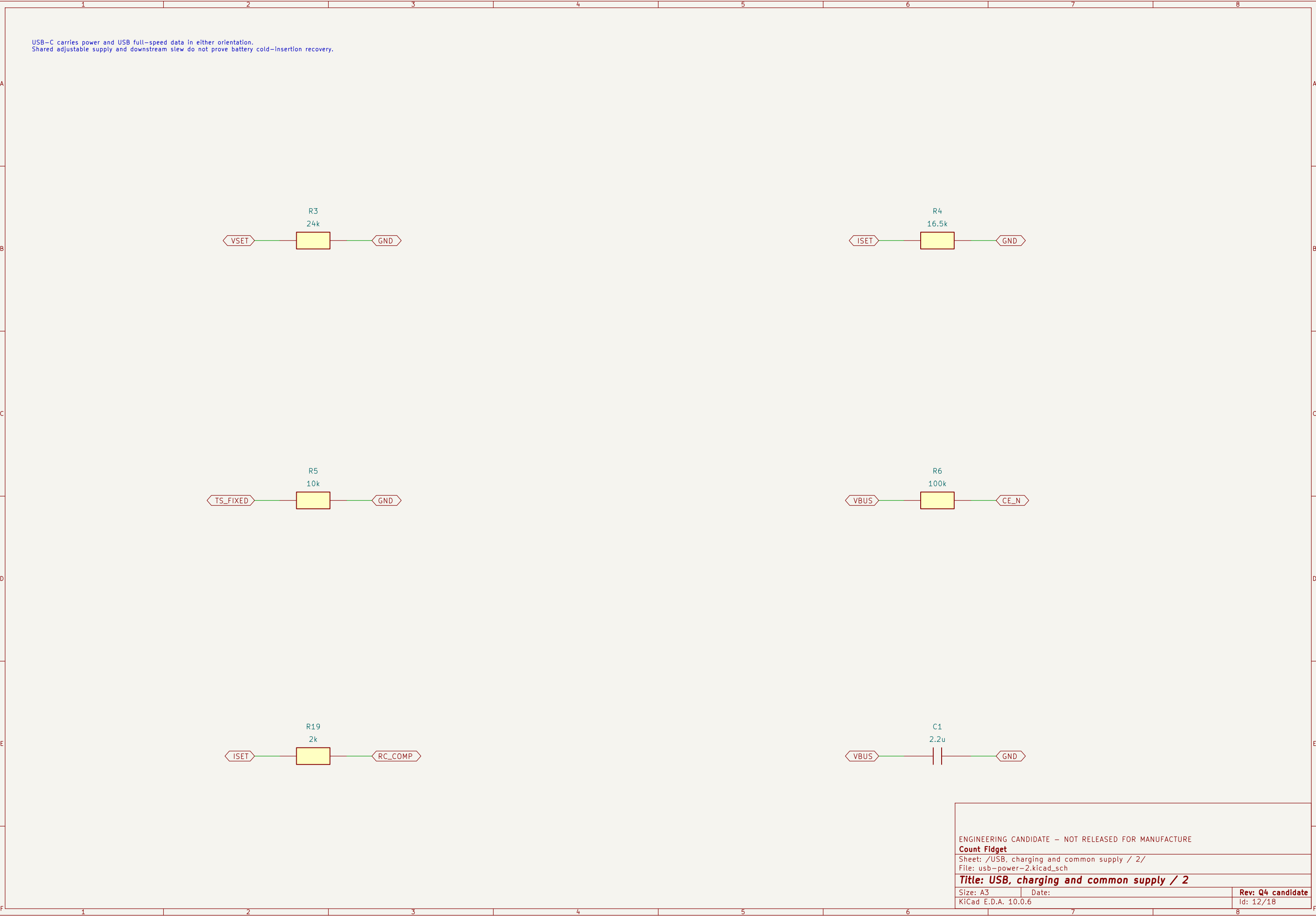


Complete HS96L01W4S03 module: seven-pin SPI interface.
Signals must be high impedance before module power is disabled. Mechanical drawing controls the footprint.

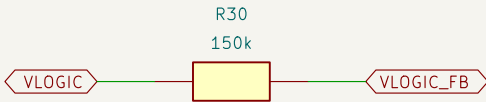
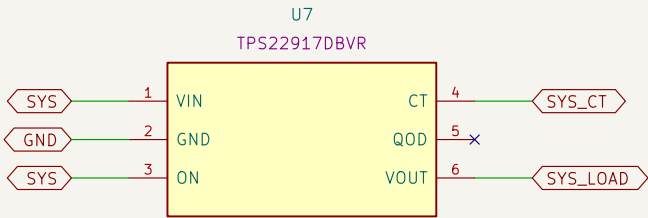
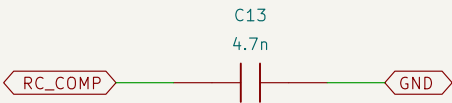








USB-C carries power and USB full-speed data in either orientation.
Shared adjustable supply and downstream slew do not prove battery cold-insertion recovery.



ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE

Count Fidget

Sheet: /USB, charging and common supply / 3/

File: usb-power-3.kicad_sch

Title: USB, charging and common supply / 3

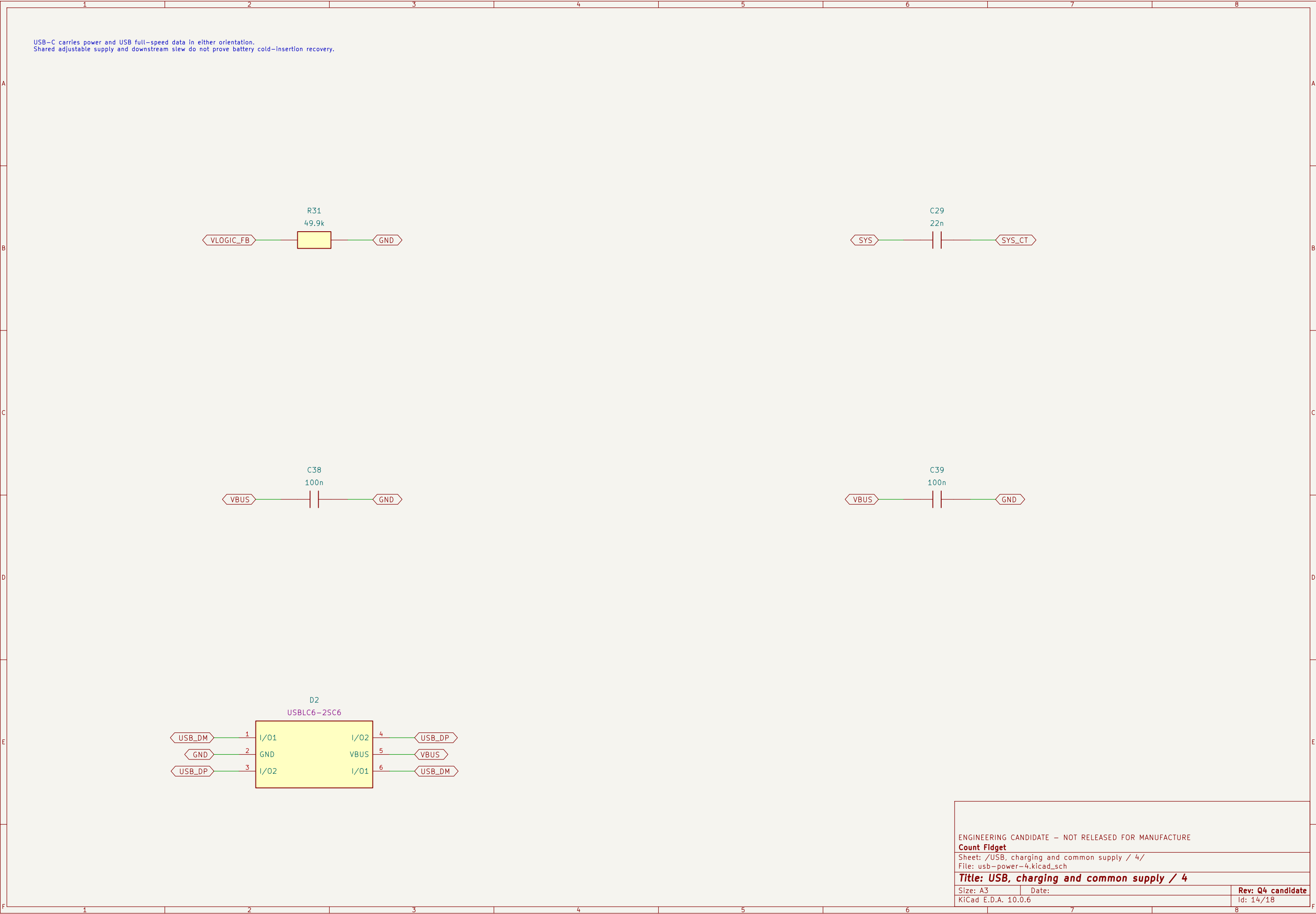
Size: A3

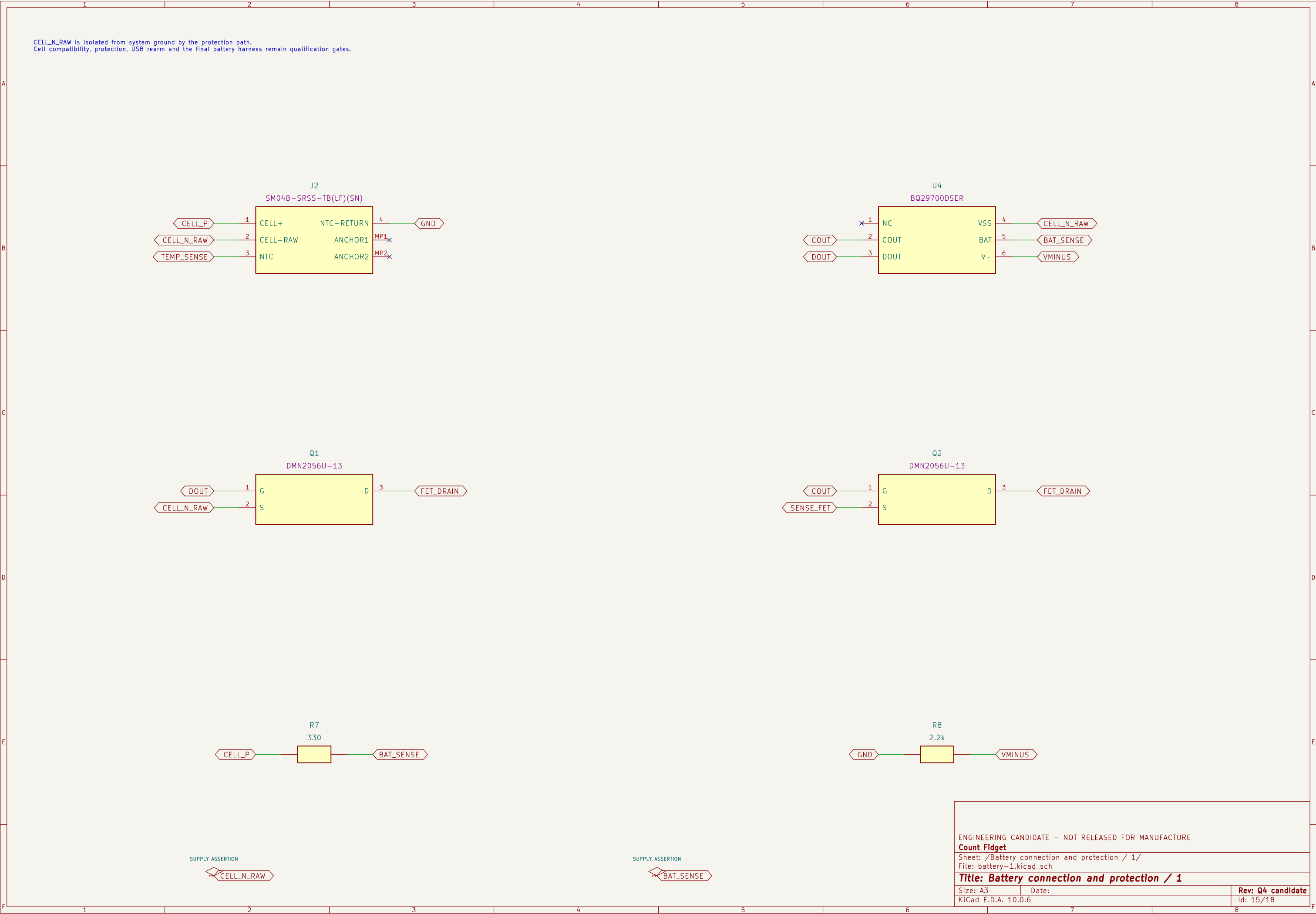
Date:

Rev: Q4 candidate

KiCad E.D.A. 10.0.6

Id: 13/18





Q1
DMN2056U-13

DOUT

1

G

CELL_N_RAW

2

S

D

3

FET_DRAIN

Q2
DMN2056U-13

COUT

1

G

SENSE_FET

2

S

D

3

FET_DRAIN

R7
330

CELL_P

BAT_SENSE

R8
2.2k

GND

VMINUS

SUPPLY ASSERTION

CELL_N_RAW

SUPPLY ASSERTION

BAT_SENSE

ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE

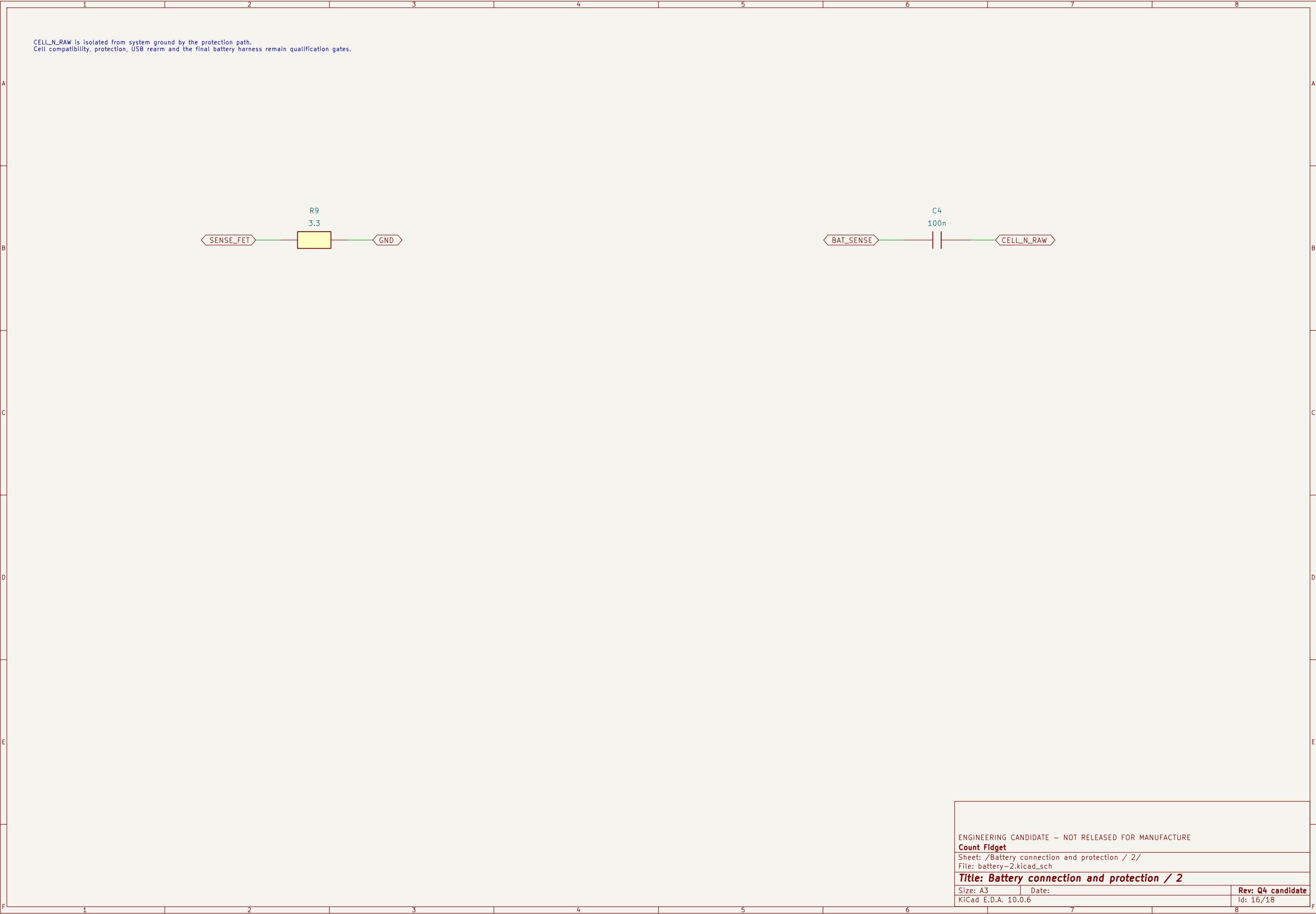
Count Fidget

Sheet: /Battery connection and protection / 1/
File: battery-1.kicad_sch

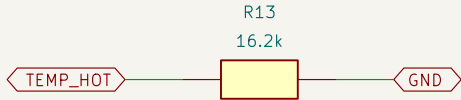
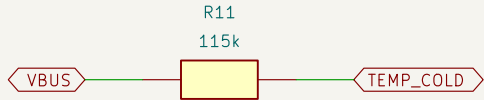
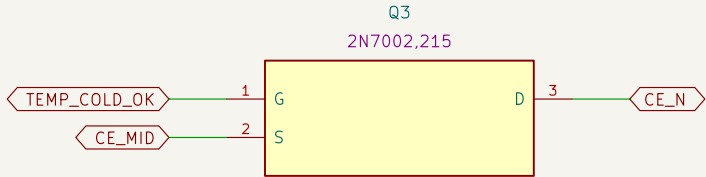
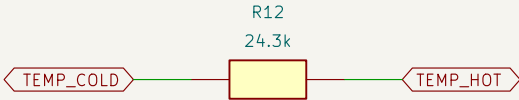
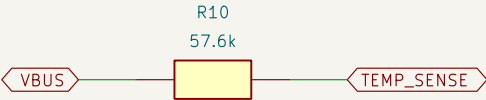
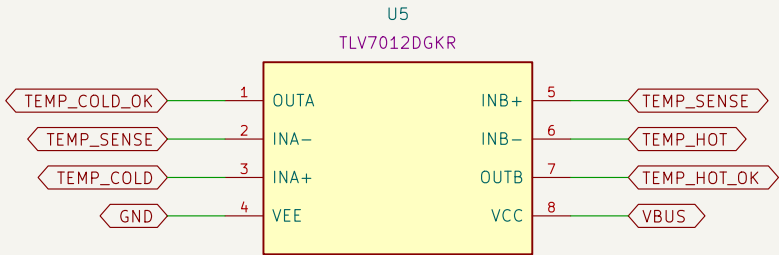
Title: **Battery connection and protection / 1**

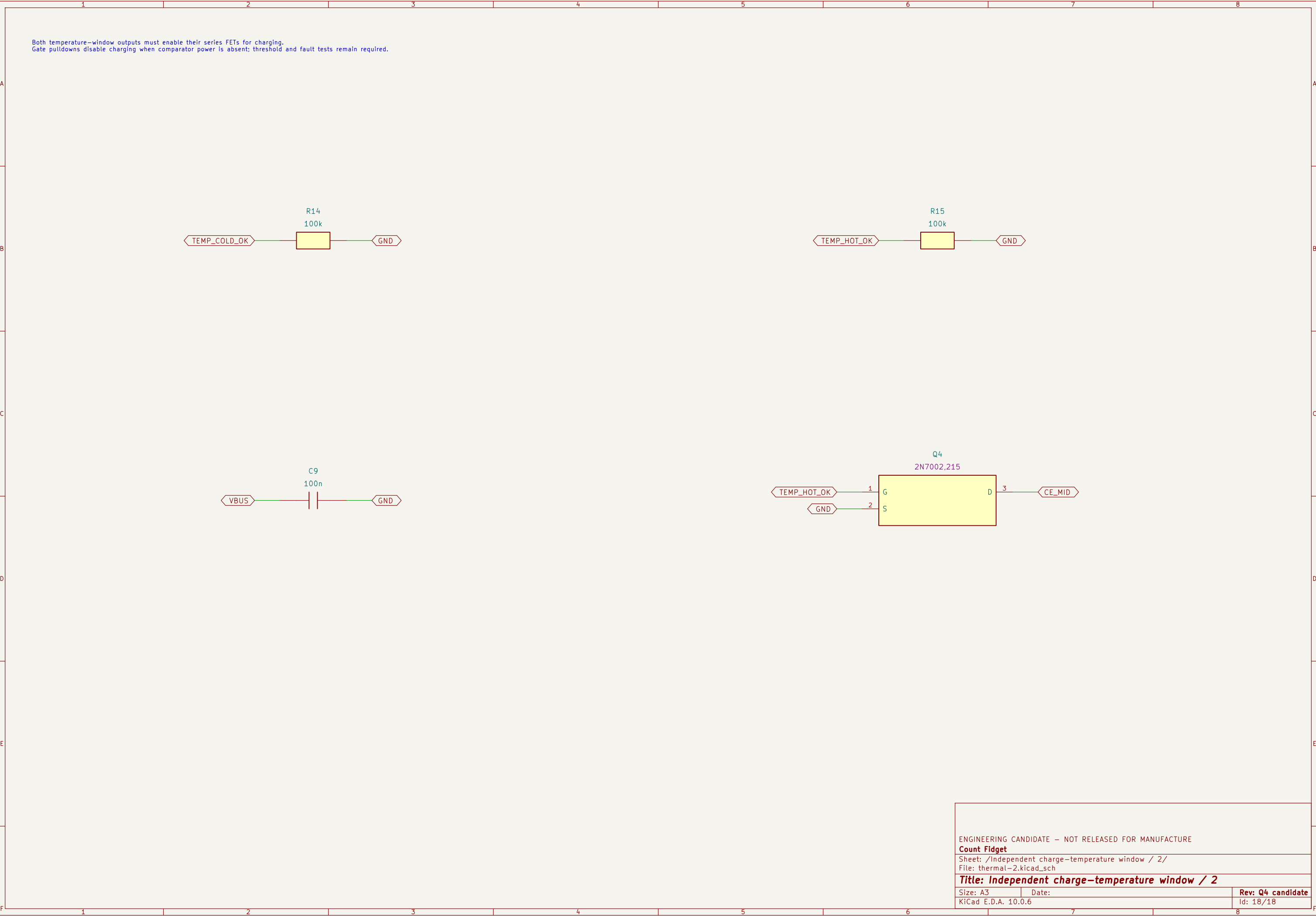
Size: A3Date:Rev: **Q4 candidate**

KiCad E.D.A. 10.0.6Id: 15/18



Both temperature-window outputs must enable their series FETs for charging.
Gate pulldowns disable charging when comparator power is absent; threshold and fault tests remain required.





ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE

Count Fidget

Sheet: /Independent charge-temperature window / 2/
File: thermal-2.kicad_sch

Title: Independent charge-temperature window / 2

Size: A3Date:Rev: Q4 candidate

KiCad E.D.A. 10.0.6Id: 18/18